

Forecasting With Temporally Aggregated Demand Signals in a Retail Supply Chain

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Suppliers of consumer packaged goods are facing an increasingly challenging situation as they work to fulfill orders from their retail partners’ distribution facilities. Traditionally these suppliers have generated forecasts of a given retailer’s orders using records of that retailer’s past orders. However, it is becoming increasingly common for retail firms to collect and share large volumes of point-of-sale (POS) data, thus presenting an alternative data signal for suppliers to use in generating forecasts. A question then arises as to which data produce the most accurate forecasts. Compounding this question is the fact that forecasters often temporally aggregate data for consolidation or to produce forecasts in larger time buckets. Extant literature prescribes two countervailing statistical effects, information loss and variance reduction, that could play significant roles in determining the impact of temporal aggregation on forecast accuracy. Utilizing a large set of paired order and POS data, this study examines these relationships.

Keywords: retail; forecasting; temporal aggregation; S&OP

INTRODUCTION

Demand planning is increasingly recognized as fundamental to efficient supply chain operations and overall firm profitability. This is particularly true for manufacturers of consumer packaged goods (CPG) in retail supply chains. CPG manufacturers are faced with the challenge of forecasting how much inventory their retailer customers will order. Since most large retailers maintain a network of distribution centers (DCs) to fulfill stores, and those DCs place orders directly to suppliers, “it is the orders placed by the retail DC (DC orders) which suppliers find difficult to forecast accurately” (Williams and Waller 2010, 231). This is because the variance of DC orders is typically much more erratic than that of consumer demand (Lee et al. 1997).

To assist with this task, many retailers now share end sales, or point-of-sale (POS) data, with suppliers, so that suppliers have access to a less variable demand signal than order history from which to generate their order forecasts (Williams and Waller 2010). Suppliers without access to POS data, or the necessary capabilities to effectively utilize POS data, frequently employ other variance reduction strategies in their demand planning processes, such as cross-sectional (e.g., geographic locations, products) demand aggregation. This results in a smoother customer demand signal, and therefore, potentially more accurate demand forecasts (Williams and Waller 2011).

Recently, the notion of temporal aggregation as a variance reduction strategy for forecasting has emerged in the supply chain management (SCM) literature (Rostami-Tabar et al. 2013). Temporal aggregation refers to the aggregation of higher frequency demand data (e.g., weekly) to lower frequency data (e.g.,

monthly). Intuitively, it seems that the lower frequency data will reduce demand variance and subsequently lead to lower forecast error. However, in this research, we contend that this intuition may be based on an incomplete theorization. Foundational to the notion that temporal aggregation will reduce forecast error is the concept of variance reduction. In this respect, temporal aggregation is similar to risk pooling. Risk pooling has been utilized in the SCM literature for decades. It underlies the portfolio effect and is known to lead to the variance reduction achieved by the consolidation of inventory holding locations (Zinn et al. 1989; Ronen 1990; Mahmoud 1992; Evers and Beier 1993, 1998; Tallon 1993; Evers 1995, 1996, 1997; Das and Tyagi 1999; Ballou 2005). Based on the notion of risk pooling, researchers have also shown in the supply chain literature that data aggregation across products (i.e., product aggregation) leads to reduced demand variance and subsequently lower forecast error (Williams and Waller 2011).

Given the recent emphasis on understanding the effect of temporal aggregation on demand data, we are motivated in this article to explore the interaction of temporal aggregation and the use of different demand signals in the task of forecasting retailer orders. In the examination of whether temporal aggregation positively or negatively affects order forecast error in the retail supply chain, we suggest that countervailing effects of temporal aggregation may exist, namely, information loss and variance reduction. The statistical theory of information loss (Amemiya and Wu 1972; Marcellino 1999) suggests that forecast error can become inflated as the underlying data generating process of the time series becomes altered and information about the underlying process may be lost as the time series is temporally aggregated. On the other hand, the concept of variance reduction suggests that random errors are cancelled as a time series is temporally aggregated, thereby reducing both the variance of the time series as well as forecast error (Finn 2004; Hotta et al. 2005).

To reconcile these competing notions, we hypothesize that the dominant effect of temporal aggregation likely depends on the

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customer demand signal being used by the supplier to create the order forecasts. That is, the decision of whether to temporally aggregate data should be dependent on whether the supplier uses shared POS data or the record of past customer orders (order data) to forecast future requirements. We suggest that this is due not only to the relative variance in the data but also to the degree of autocorrelation inherent in these demand signals. We will show that the level of first-order autocorrelation, which we later refer to simply as autocorrelation, in the data affects the degree to which variance is reduced through temporal aggregation. To test our hypotheses, we utilize a data set comprised of 2 years of weekly order and POS data for 14 different stock keeping units (SKUs) from six retail DCs placing orders to a large, CPG manufacturer located in the United States. After presenting the results of our analysis, we discuss our findings and suggest implications for both research and practice.

In the following sections, we review the forecasting and information sharing literature to set the context for our study. Hypotheses are developed from the literature bases of multiple disciplines, including logistics, econometrics, and statistics, and we analytically illustrate not only how temporal data aggregation may transform statistical properties of a time series but also reduce its variance. To test our hypotheses, we employ the data we obtained to create a forecast competition. Results, conclusions, and managerial implications are then presented and discussed.

BACKGROUND

In a retail supply chain, products are typically replenished at the store through the retailer's network of DCs. The process generally follows a model similar to that depicted in Figure 1, where a set of individual retail stores place orders to a single regional DC. Those DCs, in turn, place orders on a periodic basis to a supplier. For suppliers, it is these orders placed by the retail DCs (DC orders) that are of particular interest as they represent customer demand. (Not to be confused with consumer demand, which is measured by POS data.)

It might seem that orders could be easily forecasted by adding the POS of the stores replenished by the particular DC, however,

a number of issues introduce complexity and variability into the retailer's ordering processes. These factors, as outlined by Williams and Waller (2010), can include replenishment and execution processes at the store or operating procedures at the retail DCs (Kum et al. 2010; Vogt 2010), as well as idiosyncrasies with warehouse management systems (Autry et al. 2005). In addition, supply chain processes can drive complexity, such as postponement (Zinn and Bowersox 1988), inventory centralization (Zinn et al. 1989; Ronen 1990; Mahmoud 1992; Evers and Beier 1993, 1998; Tallon 1993; Evers 1995, 1996, 1997) and lean practices (Goldsby et al. 2006). All of these factors may make DC orders more difficult to predict, than retail sales.

To forecast orders for each SKU, suppliers often use relatively simple processes. Usually, time-series forecasting methods, such as exponential smoothing, are used to statistically predict future customer demand, using historical order data (Williams and Waller 2010). Recently, however, suppliers have begun to reexamine their customer demand planning processes and, as noted previously, are now attempting to incorporate POS data into the processes.

HYPOTHESIS DEVELOPMENT

In this research, we examine the effect of nonoverlapping temporal aggregation on order forecast error. Nonoverlapping temporal aggregation occurs when consecutive, nonoverlapping time-series data are summed to represent a larger time unit. As previously discussed, countervailing effects of temporal aggregation on forecast error seem to exist.

The first of the countervailing effects is information loss. Amemiya and Wu (1972) show the existence of information loss in an autoregressive time series. The literature later generalizes the effect of temporal aggregation to include time-series models such as autoregressive moving average models with exogenous variables (ARMAX) (Brewer 1973), seasonal structures (Wei 1978), and nonstationary models (Tiao 1972), among others. Information loss occurs because temporal aggregation results in the transformation of a time-series properties. In Table 1, we present a brief review of analytical literature that examines temporal aggregation and the associated consequences.

Figure 1: Diagram of retail supply chain.

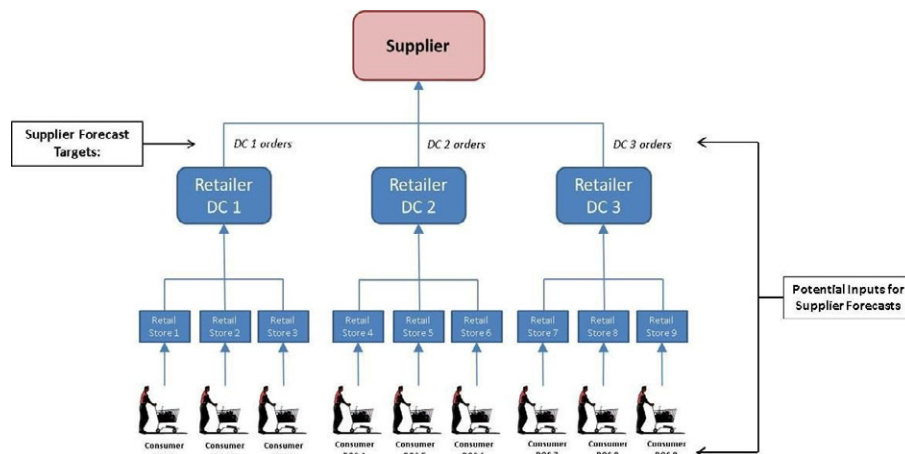


Table 1: Temporal aggregation literature: models and consequences

Author, Year	Model/Context	Findings/Implication
Amemiya and Wu 1972	AR	Rogue MA residuals structure is induced
Tiao 1972	IMA(d,q)	Short-term forecast accuracy decreases
Brewer 1973	ARMA; ARMAX	Generalized results from Amemiya and Wu (1972)
Wei 1978	Seasonal ARIMA	Seasonality parameters can be altered
Weiss 1982	ARMAX; ARIMA	Statistical causality can be lost; Induce two-way causality
Stram and Wei 1986	ARIMA	Autoregressive order can be reduced
Drost and Nijman 1993	GARCH	Parameters can be transformed
Rossana and Seater 1995	ARIMA	Short-term cyclical variations are lost
Koreisha and Fang 2004	Generalized ARMA	Short-term forecast accuracy decreases
Silvestrini and Veredas 2008	ARIMAX; ARMA-GARCH	Additional rogue error term; Parameters can be transformed

The consequences of temporal aggregation can be summarized into two categories. First, temporal aggregation can induce rogue statistical properties at the aggregate level, such as moving average residuals, statistical causality, additional error terms, and cyclical artifacts. Second, temporal aggregation can result in lost or altered time-series properties such as seasonality, autoregressive orders, short-term cyclical variations, and other parameters. A common conclusion reached by this literature is that temporal aggregation may increase forecast error due to information loss about the true underlying properties of the time series.

Despite the abundant analytical evidence in the statistics and economics literatures of information loss, demand planners often still prefer to forecast using temporally aggregated data (Finn 2004; Nikolopoulos et al. 2011). Rossana and Seater (1995) infer this may be the case because statistical theory “may not be definitive because some of the results are asymptotic and leave open the question of what happens with actual data” (p. 443). On the basis of the SCM literature, we argue that a countervailing effect to information loss exists, namely the variance reduction effect of temporal aggregation.

According to the SCM literature, aggregation tends to reduce aggregate variance. The square root law was developed to show that location-based aggregation reduces safety stock due to reduced variance (Maister 1976). The square root law was later

shown to be a special case of a broader portfolio effect, which prescribes similar benefits to aggregation under more generalized conditions (Zinn et al. 1989). Subsequent scholars expanded the portfolio effect to consider multiple consolidation points (Evers and Beier 1993), lead time (Tallon 1993; Evers and Beier 1998), cycle stock (Evers 1995), transshipments (Evers 1996), and partial aggregation (Das and Tyagi 1999). More recently, the variance reduction effect of aggregation has been observed in forecasting for both product-location (Williams and Waller 2011) as well as temporal (Finn 2004; Nikolopoulos et al. 2011; Rostami-Tabar et al. 2013) dimensions. We present in Table 2 an overview of the risk-pooling literature in SCM research.

In summary, while temporal aggregation may impede forecast accuracy through the loss of information, it may also improve forecast accuracy due to variance reduction. Figure 2 depicts this trade-off.

To illustrate both the information loss and variance reduction effects, we consider the following case where a weekly series X_t is aggregated into a nonoverlapping time series, such that $Z_t = X_t + X_{t-1}$.

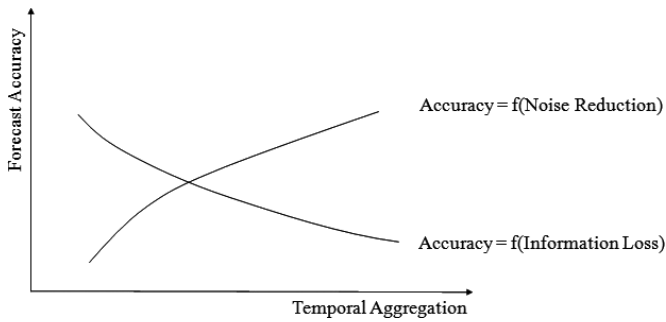
We begin by defining X_t as a generalized first-order autoregressive series, with its error term correlated over time (i.e., $\varepsilon_{1,t} + \varepsilon_{1,t-1} = \theta\varepsilon_{2,T}$) and homoskedastic (i.e., $\sigma_t = \sigma_{t-1}$). $X_t \sim \text{AR}(1)$

Table 2: Aggregation literature: logistical theory and proposed consequences

Author, Year	Model/Context	Findings/Implications
Zinn et al. 1989	Portfolio effect	Aggregation results in risk pooling to reduce demand variance
Mahmoud 1992	Portfolio quantity effect	Benefits of risk pooling can be quantified
Evers and Beier 1993	Expanded portfolio effect	Risk pooling can be managed through multiple consolidation points
Tallon 1993	Enhanced portfolio effect	Risk-pooling policies should be selective and consider lead time variability
Evers 1995	Portfolio effect	Risk-pooling benefits both safety and cycle stocks
Evers 1996	Portfolio effect	Transshipment can be used to reduce safety stock in addition to risk pooling
Evers and Beier 1998	Multiple models	Consolidation point's lead time influences risk-pooling effect
Das and Tyagi 1999	Portfolio effect	Risk pooling may benefit even partial aggregation
Finn 2004	Exponential smoothing	Risk-pooling results in more accurate aggregate forecasting
Nikolopoulos et al. 2011	ADIDA	Temporal aggregation can reduce demand variance for intermittent demand
Williams and Waller 2011	Demand planning	Product-location aggregation may be selectively used to improve forecast
Rostami-Tabar et al. 2013	AR(1)	Temporal aggregation may be selectively used to improve forecast

Note: ADIDA, Aggregate-Disaggregate Intermittent Demand Approach.

Figure 2: Trade-off between information loss and variance reduction.



can be expressed as $X_t = \varphi_1 X_{t-1} + \mu_1 + \varepsilon_{1,t}$, where $\varepsilon_{1,t} \sim N(0, \sigma_1^2)$, μ_1 is the constant, and φ_1 is the first-order autoregressive factor. According to Brewer (1973), the aggregated series Z_t can be expressed as $Z_t = \varphi_2 z_{t-1} + \mu_2 + \theta \varepsilon_{2,T}$, where $\varepsilon_{2,T} \sim (0, \sigma_2^2)$, φ_2 and μ_2 are the first-order autoregressive factor and the constant average of Z_t , respectively. θ is the factor of change in the error term over time, which is denoted by $\varepsilon_{2,T}$ and has a variance of σ_2 . Furthermore, whereas the expected value and variance for X_t are known to be $E[X_t] = \frac{\mu_1}{1-\varphi_1}$ and $Var[X_t] = \frac{\sigma_1^2}{1-\varphi_1^2}$, they are instead $E[Z_t] = \frac{\mu_2}{1-\varphi_2}$, and $Var[Z_t] = \frac{\sigma_2^2(1+\theta^2-2\varphi_2\theta)}{1-\varphi_2^2}$ for Z_t (Tiao 1972). In other words, temporal aggregation results in the transformation of X_t into an entirely different data generation process (Z_t).¹ Whereas X_t is AR(1), Z_t is ARMA(1,1) and has parameters independent of X_t .

Next, we examine temporal aggregation's variance reduction effect on the standard deviation of the time series. We again consider $X_t \sim AR(1)$, for which σ_t represents the standard deviation of the time series at time t , while σ_{t-1} represents the standard deviation of X_{t-1} . Variance reduction effect can be observed if the standard deviation of the aggregated time series, Z_t , is lower than twice that of X_t . That is, $\sigma_2 < 2\sigma_1$.

The standard deviation of Z_t can be expressed² as $\sigma_2 = \sqrt{\sigma_{1,t}^2 + \sigma_{1,t-1}^2 + 2\varphi\sigma_{1,t}\sigma_{1,t-1}}$. Since $\sigma_{1,t} = \sigma_{1,t-1}$ due to constant variance, the variance of the temporally aggregated time series can be rewritten as, $\sigma_2 = \sqrt{\sigma_{1,t}^2 + \sigma_{1,t}^2 + 2\varphi\sigma_{1,t}\sigma_{1,t}}$ and can be simplified algebraically as, $\sigma_2 = \sigma_1\sqrt{2(1+\varphi)}$, indicating the variance of the temporally aggregated time series is reduced unless the time series is perfectly, positively autocorrelated, that is, $\varphi = 1$.

A major contributing factor to order forecast error is the bullwhip effect. The bullwhip effect is defined as the amplification

¹According to Tiao (1972), φ_1 , φ_2 , and θ are independent of each other.

²According to Zinn et al. (1989), product-location aggregation yields standard deviation of demand that may be expressed as $\sqrt{\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2}$, where σ_1 and σ_2 are the standard deviations of demand for two demand series with their correlation measured as ρ . In a temporal aggregation setting, σ_t and σ_{t-1} , with φ being their degree of autocorrelation, are substituted for σ_1 , σ_2 , and ρ , respectively.

of demand variance as orders move from the retail echelon to the manufacturing echelon of the supply chain (Lee et al. 1997). As retail sales translate into orders placed by stores to the retail DC, and then on to a supplier's DC, the variance of orders is amplified at each echelon, resulting in a more "noisy" demand signal. As shown above, temporal aggregation can be used to reduce the variance of demand signals. However, the SCM literature prescribes that POS should be shared and used to forecast orders for this same purpose.

As POS data generally contains less variance than order data, we hypothesize that the net effect of temporal aggregation on these demand signals may differ given the countervailing effects of information loss and variance reduction. Given that the bullwhip effect amplifies variance of order data, we expect that the variance reduction effect of temporal aggregation may be the dominant of the two effects and lead to lower order forecast error, when order data are the demand signal used to generate the order forecasts.

On the contrary, POS data are not subject to the bullwhip effect and has lower variance than order data. Therefore, the potential benefit of variance reduction due to temporal aggregation is much less. In fact, we contend that the information loss effect may be dominant when temporally aggregating POS data to generate order forecasts. That is, as POS data are temporally aggregated, the loss of information about the underlying nature of sales has a negative effect on the ability to forecast orders and overshadows any potential benefit of variance reduction. Thus, we hypothesize:

H₁: When order data is used to generate forecasts, temporal aggregation is negatively associated with order forecast error.

H₂: When POS data is used to generate the forecasts, temporal aggregation is positively associated with order forecast error.

We further suggest that the effects of temporal aggregation on order forecast error may be moderated by the degree of autocorrelation in the given demand signal. The moderating effect of autocorrelation is related to the variance reduction effect of temporal aggregation identified earlier. Recall from the variance reduction discussion that if $\varphi = 1$ (i.e., perfect positive autocorrelation), then $\sigma_2 = 2\sigma_1$. Otherwise, for all $\varphi < 1$ (i.e., less than perfect positive autocorrelation), then $\sigma_2 < 2\sigma_1$. In addition, we also observe that $\sigma_2 \rightarrow 2\sigma_1$, as $\varphi \rightarrow 1$. That is, we observe that variance reduction due to temporal aggregation decreases as the autocorrelation increases. Therefore, we expect that as autocorrelation increases, the variance reduction effect will weaken and the information loss will be relatively stronger. Thus, we hypothesize that autocorrelation moderates the effect of temporal aggregation on order forecast error when either demand signal is used to generate the forecasts.

H_{3a}: When order data is used to generate forecasts, the effect of temporal aggregation on forecast error is moderated by autocorrelation.

H_{3b}: When POS data is used to generate forecasts, the effect of temporal aggregation on forecast error is moderated by autocorrelation.

METHODS AND MEASURES

Demand forecast competition design

While we maintain that the consideration of the countervailing effects of variance reduction and information loss among various demand streams is an important contribution of this article, we also recognize that the complexity of order forecasting processes requires that additional factors be considered if this work is to have significant managerial implications. Specifically, a thorough consideration must address the fact that retailer order forecasts are required in multiple time “buckets” (i.e., levels of time).

Suppliers face the challenge of generating order forecasts over a given horizon at multiple levels of time because of the different needs within the firm. For example, order forecasts may be used for inventory and transportation deployment to retail customer DCs. These forecasts signal where and when to stage inventory within the supplier’s distribution network to fulfill its customers’ orders and the transportation capacity that will be required at each node in the distribution network. Given that shipments to customers often occurs on a weekly (or even more frequent) basis, weekly order forecasts are needed by the supplier. However, other activities may require a monthly view of customer orders. For example, many CPG suppliers use order forecasts to inform corporate-level production, sales, and marketing planning processes. For this type of purpose, suppliers may generate order forecasts at a monthly level as well.

To test our hypotheses, we design a forecast competition simulating weekly and monthly forecast error based on 2 years of weekly retail DC order and POS data obtained from a large, U.S. CPG manufacturer. For both weekly and monthly forecasts, the competition compares the errors between forecasts generated using order data and POS data, taking both a nonaggregated approach (i.e., forecasts generated using weekly data), and a temporally aggregated approach (i.e., forecasts generated using data aggregated into monthly buckets). Table 3 helps make salient the number of resulting forecast approaches that must be examined, and thus the scope of this research.

Our data were collected from two commonly shopped, nonseasonal grocery categories. The first category is a mature, dry grocery product that is one of the highest volume categories in the industry. The second category features fresh, refrigerated products that have short shelf-lives. Our sample includes weekly data for nine dry grocery SKUs and five refrigerated SKUs. The data

were collected over a period of 104 weeks at six regional U.S. DCs owned by one of the manufacturer’s largest retail customers, for a total of 82 unique SKU-DC combinations.³ DC orders are defined as the weekly orders for a given SKU placed by a particular retail DC to the manufacturer, while POS is the cumulative weekly sales of a given SKU at the retail stores replenished by the particular DC.

For short-term forecasts of fast-moving consumer goods items, exponential smoothing models are the most commonly utilized in practice (Mentzer and Kahn 1995; McCarthy et al. 2006). These models are most likely popular because they are simple to use (Gardner 1985) and can be easily adjusted to respond to changes in the data being forecasted. Furthermore, they generally offer forecast accuracy competitive against other approaches that are substantially more complex (Makridakis et al. 1982; Makridakis and Hibon 2000). Considering that trend appears to be present in the data, for which exponential smoothing alone is not sufficient, we utilize Holt’s exponential smoothing model, which specifically accounts for trend, to generate the statistical forecasts.

Weekly forecasts

We forecast weekly orders taking a nonaggregated approach, using both order and POS data. The in-sample period of our data contains 91 weekly observations (21 months of weekly data). During this in-sample period, the level and trend components are estimated using the trend-adjusted exponential smoothing parameters. In our approach, multiple sets of smoothing parameters are used to forecast a given SKU-DC combination. This approach is chosen to reduce the possible influence of a single parameter choice on our results. Smoothing parameters are selected in two ways. We refer to the first group of smoothing parameters as selected parameters, where three values for α and β ($\alpha = 0.51, 0.19, 0.02$; $\beta = 0.176, 0.053, 0.005$) are selected based upon the range of reasonable values given by Silver et al. (1998, 108). Of the nine possible combinations of these reasonable parameters, we utilized six sets of parameters for which the values of β are well below those of α due to forecast stability (McClain and Thomas 1973). We refer to the second group of smoothing parameters as optimized parameters, where the α and β values were found through an optimization which minimized in-sample forecast error.

Irrespective of the smoothing parameters, the initial forecast was set to the value of the actual order for the first period (Hauke and Wichern 2005, 118); this is referred to as forecast initialization. We then utilize the level and trend components estimated during the in-sample period to generate order forecasts for each SKU-DC combination over the 13 weeks out-of-sample forecast horizon (three-months) using both groups of smoothing parameters. For each SKU-DC combination, this procedure is conducted where POS data are the demand signal used to forecast orders and where order data are used to forecast orders.

We also generate weekly forecasts using an aggregated approach. First, 91 weekly demand observations are aggregated into 21 nonoverlapping monthly demand observations. During

Table 3: Forecast approaches

Forecast time bucket	Temporal approach	Demand signal
Weekly	Nonaggregated	Order data POS data
	Aggregated	Order data POS data
Monthly	Nonaggregated	Order data POS data
	Aggregated	Order data POS data

³We note that one SKU was not stocked at two DCs, hence we have 82, rather than 84 SKU-DC combinations.

the in-sample period, the level and trend components are estimated based on the aggregated data using both groups of smoothing parameters. The smoothed level and trend components of monthly demand are estimated for both orders and POS. Next, the estimated level and trend components are used to generate demand forecasts for the three-month out-of-sample period. Finally, because the resultant forecasts are at the monthly level, they are disaggregated into weekly forecasts by dividing the monthly forecast into the number of weeks in the month.

Monthly forecasts

As with weekly forecasts, monthly forecasts are generated using both the nonaggregated and aggregated temporal approaches. Taking the nonaggregated approach, we use the 91 weekly observations of order and POS data to estimate level and trend components, and then produce forecasts, using the procedure described for weekly forecasts, for a 13-week out-of-sample horizon. Those forecasts are summed into nonoverlapping monthly buckets to produce monthly forecasts.⁴ For the aggregated approach, we use the data which were aggregated into 21 monthly observations. We then generate monthly forecasts for each SKU-DC combination over a three-month out-of-sample forecast horizon.

Given that each of the three dimensions of the forecast competition (forecast time bucket, temporal approach, and demand signal) contains two levels, eight (2^3) different forecast approaches were utilized. The procedures for each method are summarized in Table 4.

Finally, we utilized the six sets of selected forecast parameters and optimized parameters for Holt's exponential smoothing model to generate both weekly and monthly customer demand forecasts using two sources of demand signals (POS and order data).

ANALYSIS AND RESULTS

Forecast competition results

To initially evaluate the forecast error for each of the eight forecast approaches, we calculate mean absolute deviation (MAD) over the forecast horizon. MAD measures forecast error by averaging the absolute value of the forecast errors over the forecast horizon. It is also one of the most frequently used measures of forecast errors in practice (Mentzer and Kahn 1995).⁵ Next, we split the sample based on forecast type (i.e., weekly and monthly forecasts). For each sample, we compare MAD across demand

signals and temporal approaches with a one-way analysis of variance. Results are shown in Table 5.

For weekly forecasts when using a nonaggregated approach, POS-based forecast error is significantly lower than order-based forecast error (132.73 vs. 147.30, $F_{1, 14,923} = 33.08$, $p < 0.01$) but is only nominally lower when an aggregated approach is used (140.89 vs. 141.05, $F_{1, 14,923} = 0.04$, $p > 0.10$).

Similarly for monthly forecasts, the use of POS data results in significantly less forecast error than order data when a nonaggregated approach is taken (323.77 vs. 360.15, $F_{1, 3,358} = 4.38$, $p < 0.01$), but when an aggregated approach is employed order data outperforms POS data (385.97 vs. 313.86, $F_{1, 3,358} = 15.95$, $p < 0.01$).

Empirical models

To statistically test our hypotheses, we take a hierarchical linear modeling (HLM) approach. As our data are taken from 14 products nested in two categories at six DCs, observations cannot be assumed to be independent, potentially violating an underlying assumption of ordinary least squares estimation. HLM is well-suited for our data because it parcels out variance components based on nested structures in the data (Raudenbush and Bryk 2002).

Our dependent variable is MAD, where MAD_W is the weekly forecast MAD over the forecast horizon and MAD_M is the monthly forecast MAD over the forecast horizon. As we are interested in assessing the interaction of the demand signal and temporal approach used to forecast, we code each observation with respect to these two dimensions. First, *AGG* is a binary variable that denotes whether the temporal approach is aggregated (1) or nonaggregated (0). *POS* is another binary variable that denotes whether the forecast is based on POS (1) or order (0) data. Next, we obtain each SKU-DC combination's respective autocorrelation factor, *AF*, for both its POS and order data by estimating its autocorrelation coefficient with in-sample demand observations. In addition, we include the average demand, *MEAN*, for each product corresponding to the temporal approach of the forecast (Mentzer and Cox 1984). This notation, as well notation used in the following sections, is listed for reference in Table 6.

For both weekly and monthly forecast errors, we test our hypotheses regarding the interaction of the temporal approach, demand signal, and autocorrelation in multiple steps. Similar to the approach of Ang et al. (2002), DeHoratius and Raman (2008), and Liao and Chuang (2004), we estimate our empirical models using full maximum likelihood in three stages. First, we estimate a null model where no control or predictor variables are included. In Model 1, we add control variables and the direct effects of our hypothesized interactions. In Model 2, the two-way interaction term between *POS* and *AGG* is added to test H_1 and H_2 . Finally, in Model 3, we add the remaining two-way interaction terms and the three-way interaction term to test H_{3a} and H_{3b} .

Null models

Following established HLM procedure, we begin by partitioning both weekly ($MAD_{W,ijk}$) and monthly forecast errors ($MAD_{M,ijk}$) into variance components across products i ($i = 1, \dots, 14$),

⁴Note that in aggregating weekly forecasts to monthly equivalents, we combined weeks 1–4 to form month 1, weeks 5–8 for month 2, and weeks 9–13 for month 3.

⁵We also estimated forecast error as mean squared error and performed all subsequent analyses with this alternative measure. All results remained qualitatively unchanged. Therefore, we present only the results obtained with MAD to be parsimonious.

Table 4: Forecast procedures

Forecast time bucket	Temporal approach	Demand signal	Procedure
Weekly	Nonaggregated	Order data	Use weekly order data to generate weekly forecasts
		POS data	Use weekly POS data to generate weekly forecasts
	Aggregated	Order data	Aggregate weekly order data into monthly buckets; generate monthly statistical forecasts; disaggregate monthly forecasts into weekly forecasts
		POS data	Aggregate weekly POS data into monthly buckets; generate monthly statistical forecasts; disaggregate monthly forecasts into weekly forecasts
Monthly	Nonaggregated	Order data	Use weekly order data to generate weekly statistical forecasts; sum weekly forecasts into monthly forecasts
		POS data	Use weekly POS data to generate weekly statistical forecasts; sum weekly forecasts into monthly forecasts
	Aggregated	Order data	Aggregate weekly order data into monthly buckets; generate monthly statistical forecasts
		POS data	Aggregate weekly POS data into monthly buckets; generate monthly statistical forecasts

Table 5: Mean absolute deviation

Forecast time bucket	Temporal approach	Demand signal	MAD	F-value
Weekly	Nonaggregated	Order data	147.30	33.08***
		POS data	132.73	
	Aggregated	Order data	141.05	0.04
		POS data	140.89	
Monthly	Nonaggregated	Order data	360.15	4.38**
		POS data	323.77	
	Aggregated	Order data	313.86	15.95***
		POS data	385.97	

** $p < 0.05$, *** $p < 0.01$.

categories j ($j = 1, 2$), DCs k ($k = 1, \dots, 6$), and combinations of forecast parameter f ($f = 1, \dots, 6$). Thus, our null models are:

$$MAD_{W,ijkf} = \theta_{W,0} + CAT_{W,000j} + DC_{W,000k} + FP_{W,000f} + e_{W,ijkf}$$

$$MAD_{M,ijkf} = \theta_{M,0} + CAT_{M,000j} + DC_{M,000k} + FP_{M,000f} + e_{M,ijkf}$$

For both models, θ_0 is the fixed intercept parameter, while CAT_{000j} , DC_{000k} , and FP_{000f} are the random effect parameters normally distributed around a zero mean and variances of τ_{CAT000} , τ_{DC000} , τ_{FP000} , and σ^2 , respectively. Partitioned variance components (Null Model in Tables 7 and 8) indicate that 12% of $MAD_{W,ijkf}$ ($\sigma_W^2 = 2,837.94$, $p < 0.01$) and 48% of $MAD_{M,ijkf}$ ($\sigma_M^2 = 125,510.35$, $p < 0.01$) may be attributed to product level effects.

Conditional models

For both $MAD_{W,ijkf}$ and $MAD_{M,ijkf}$, we hierarchically enter the fixed direct and interaction effects of the independent variables $MEAN_{ijkf}$, AF_{ijkf} , AGG_{ijkf} , and POS_{ijkf} to their respective null models (Tables 7 and 8). Conditional Model 1 includes all direct effects. Model 2 includes $AGG_{ijkf} * POS_{ijkf}$, which allows us to

Table 6: Variable notation

Variable	Notation	Type	Definition
Dependent	MAD_{ijkf}	Continuous	Mean absolute deviation of forecast error for i, j, k, f
Fixed effect	θ_0	Continuous	Intercept
Random effect	CAT_{000j}	Categorical	Product category j
	DC_{000k}	Categorical	Distribution center k
	FP_{000f}	Categorical	Forecast parameter combination f
Independent	e_{ijkf}	Continuous	Error for i, j, k, f
	$MEAN_{ijkf}$	Continuous	Average demand for i, j, k, f
	AF_{ijkf}	Continuous	First-order autoregressive term for i, j, k, f
	AGG_{ijkf}	Categorical	Aggregated forecast approach for i, j, k, f
	POS_{ijkf}	Categorical	Use of point-of-sale data for i, j, k, f

assess H_1 and H_2 . Finally, in Model 3 we include the full factorial of two-way interaction effects for AF_{ijkf} , AGG_{ijkf} , and POS_{ijkf} as well as their three-way interaction to test the moderating influence of AF_{ijkf} (H_{3a} and H_{3b}). Thus, the full conditional models are specified as

$$\begin{aligned}
 MAD_{W,ijkf} = & \theta_{W,0} + CAT_{W,000j} + DC_{W,000k} + FP_{W,000f} + e_{W,ijkf} \\
 & + \pi_{W,1}MEAN_{W,ijkf} + \pi_{W,2}AF_{W,ijkf} + \pi_{W,3}AGG_{W,ijkf} \\
 & + \pi_{W,4}POS_{W,ijkf} + \pi_{W,5}AGG_{W,ijkf} * POS_{W,ijkf} \\
 & + \pi_{W,6}POS_{W,ijkf} * AF_{W,ijkf} + \pi_{W,7}AGG_{W,ijkf} * AF_{W,ijkf} \\
 & + \pi_{W,8}AGG_{W,ijkf} * POS_{W,ijkf} * AF_{W,ijkf}
 \end{aligned}$$

Table 7: Hierarchical linear modeling results for weekly forecasts

Parameter estimates	Null	Model 1	Model 2	Model 3
Fixed effects				
<i>Intercept</i>	139.45 (35.37)***	140.62 (34.26)***	144.48 (34.24)***	143.30 (34.48)***
$\pi_{W,1}$ <i>MEAN</i>		0.01 (0.00)***	0.01 (0.00)***	0.01 (0.00)***
$\pi_{W,2}$ <i>AF</i>		0.87 (2.37)	2.24 (2.37)	−6.08 (4.42)
$\pi_{W,3}$ <i>AGG</i>		−6.07 (0.90)***	−13.53 (1.09)***	−10.23 (1.42)***
$\pi_{W,4}$ <i>POS</i>		−7.94 (2.21)***	−16.48 (2.32)***	2.99 (3.99)
$\pi_{W,5}$ <i>AGG * POS</i>			14.65 (1.23)***	−15.30 (4.61)***
$\pi_{W,6}$ <i>POS * AF</i>				−16.33 (6.96)**
$\pi_{W,7}$ <i>AGG * AF</i>				13.64 (5.08)***
$\pi_{W,8}$ <i>AGG * POS * AF</i>				24.46 (7.88)***
Random effects				
<i>CAT</i>	16,849.33 (6614.61)**	15,791.35 (6204.39)**	15,774.37 (6197.54)**	15,990.55 (6280.95)**
<i>DC</i>	4,170.29 (652.81)***	3,965.59 (621.17)***	3,957.66 (619.93)***	3,985.32 (624.27)***
<i>FP</i>	360.80 (57.19)***	360.40 (57.12)***	360.43 (57.12)***	360.51 (57.13)***
<i>Residual</i>	2,837.94 (23.30)***	2,813.72 (23.10)***	49,869.91 (943.79)***	2,795.75 (22.95)***
−2 Log Likelihood	322,947.77	322,688.37	322,546.13	322,498.80
χ^2		259.40	142.24	47.33

** $p < 0.05$, *** $p < 0.01$.

Standard errors are shown in parentheses.

$$\begin{aligned}
 MAD_{M,ijkf} = & \theta_{M,0} + CAT_{M,000j} + DC_{M,000k} + FP_{M,000f} + e_{M,ijkf} \\
 & + \pi_{M,1}MEAN_{M,ijkf} + \pi_{M,2}AF_{M,ijkf} + \pi_{M,3}AGG_{M,ijkf} \\
 & + \pi_{M,4}POS_{M,ijkf} + \pi_{M,5}AGG_{M,ijkf} * POS_{M,ijkf} \\
 & + \pi_{M,6}POS_{M,ijkf} * AF_{M,ijkf} + \pi_{M,7}AGG_{M,ijkf} * AF_{M,ijkf} \\
 & + \pi_{M,8}AGG_{M,ijkf} * POS_{M,ijkf} * AF_{M,ijkf}
 \end{aligned}$$

For both models,⁶ π_1 to π_4 are the product-level fixed effects due to direct influence from mean, autocorrelation factor, aggregation approach, and the demand signal, respectively. π_5 to π_7 are the fixed effects across products for the two-way interaction effects among AR_{ijkf} , AGG_{ijkf} , and POS_{ijkf} . Finally, π_8 is the estimated three-way interaction effect.

Model results

Results for weekly forecast model

Full HLM results for weekly forecast model are displayed in Table 7. For all models, additional direct and interaction terms offered significant explanatory power, as indicated by the χ^2 .

The coefficients of interest of this study are the two-way interaction term between *POS* and *AGG*, and the three-way interaction term for *POS*, *AGG*, and *AF*. HLM results of $AGG_{W,ijkf} * POS_{W,ijkf}$ (Table 7, Model 2) show this relationship with be statistically significant ($\pi_{W,5} = 14.65$, $p < 0.01$). Figure 3 illustrates the ordinal interaction effect between the two terms.

The simple effect analyses from the HLM results show that the aggregate approach has opposing effects on forecast accuracy

when different sources of demand signal is utilized. When using order data (i.e., $POS_{W,ijkf} = 0$), an aggregated approach significantly decreases $MAD_{W,ijkf}$ (147.3 vs. 140.89, $F_{4, 14,920} = 10.47$, $p < 0.01$) in support of **H₁**. On the other hand, when POS data are utilized (i.e., $POS_{W,ijkf} = 1$), an aggregated approach significantly increases $MAD_{W,ijkf}$ (132.73 vs. 141.05, $F_{4, 14,920} = 5.75$, $p < 0.01$). Thus, we also find support for **H₂**.

Results from Model 3 (Table 7) show the three-way interaction of interest, $AGG_{W,ijkf} * POS_{W,ijkf} * AF_{W,ijkf}$, to be statistically significant ($\pi_{W,8} = 24.46$, $p < 0.01$). Figure 4 illustrates the interaction hypothesized by **H₃**.

For both order and POS data, results clearly show the impact of the autocorrelation factor on the relationship between temporal approaches and $MAD_{W,ijkf}$ to support hypotheses **H_{3a}** and **H_{3b}**. Note that we measured $AF_{W,ijkf}$ as a continuous variable with theoretical minimum and maximum values of -1 and 1 , respectively. Hence, both graphs in Figure 4 are anchored by -1 and 1 .

H₃ hypothesizes that aggregation affects $MAD_{W,ijkf}$ based on both demand signals in the same direction, but for different reasons. **H_{3a}** predicts that when order data are used, the variance reduction effect of the aggregated approach on $MAD_{W,ijkf}$ is reduced as $AF_{W,ijkf}$ becomes increasingly positive. In addition, **H_{3b}** predicts that when POS data are used, the information loss effect of the aggregated approach on $MAD_{W,ijkf}$ is amplified as $AF_{W,ijkf}$ becomes increasingly positive. In both cases, the net impact of the aggregated approach is increased forecast errors. Figure 4 shows this to be the case as the lines associated with both aggregated order and POS data have a positive slope as autocorrelation moves from negative to positive.

Results for monthly forecasts

As in the analysis of weekly forecasts, the coefficients of interest for monthly forecasts are the two-way interaction term between *AGG* and *POS*, and the three-way interaction term for *AGG*,

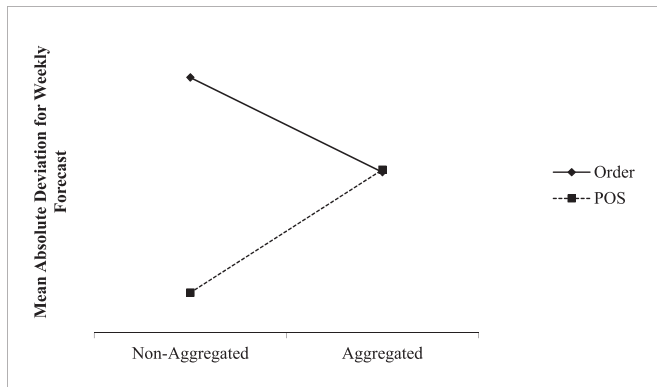
⁶Note that the estimated fixed effects are differentiated between weekly and monthly forecast errors with subscripts *W* and *M*, respectively.

Table 8: Hierarchical linear modeling results for monthly forecasts

Parameter estimates	Null	Model 1	Model 2	Model 3
Fixed effects				
<i>Intercept</i>	341.82 (75.91)***	−17.64 (33.48)	20.12 (33.99)	11.04 (34.91)
$\pi_{M,1}$ <i>MEAN</i>		0.25 (0.02)***	0.25 (0.02)***	0.25 (0.02)***
$\pi_{M,2}$ <i>AF</i>		−81.44 (23.86)***	2.83 (28.63)	44.59 (56.90)
$\pi_{M,3}$ <i>AGG</i>		22.66 (9.64)**	−47.38 (16.36)***	−32.01 (16.11)**
$\pi_{M,4}$ <i>POS</i>		75.56 (18.98)***	−38.95 (28.75)	−14.71 (54.05)
$\pi_{M,5}$ <i>AGG * POS</i>			109.63 (20.72)***	−60.85 (63.92)
$\pi_{M,6}$ <i>POS * AF</i>				−120.74 (91.65)
$\pi_{M,7}$ <i>AGG * AF</i>				13.47 (66.79)
$\pi_{M,8}$ <i>AGG * POS * AF</i>				203.45 (100.67)**
Random effects				
<i>CAT</i>	69,569.24 (30423.27)**	724.32 (3025.11)	736.89 (3051.83)	913.73 (2981.59)
<i>DC</i>	50,659.52 (8942.87)***	27,840.66 (4781.49)***	27,266.04 (4691.92)***	27,075.79 (4670.94)***
<i>FP</i>	14,809.11 (2563.11)***	14,740.46 (2520.74)***	14,762.78 (2523.97)***	14,800.12 (2532.57)***
<i>Residual</i>	125,510.35 (2192.42)***	125,160.42 (2186.30)***	124,656.40 (2177.48)***	124,436.71 (2173.65)***
−2 Log Likelihood	98,468.91	98,377.23	98,349.32	98,337.41
χ^2		91.68	27.91	11.91

** $p < 0.05$, *** $p < 0.01$.

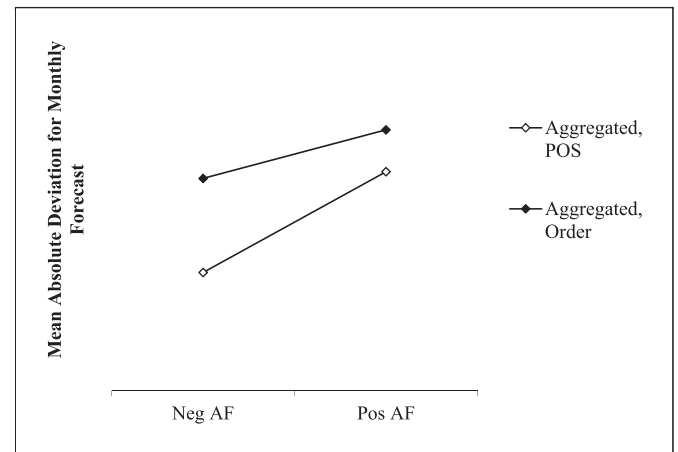
Standard errors are shown in parentheses.

Figure 3: Interaction plot of temporal aggregation and demand signal for weekly forecast error.

POS, and *AF*. We first examine $AGG_{M,ijkf} * POS_{M,ijkf}$. HLM results show this relationship with be statistically significant ($\pi_{M,5} = 109.63$, $p < 0.01$; see Table 8, Model 2). Figure 5 illustrates this significant interaction.

Simple effects analyses of the marginal means obtained from HLM indicate results similar to weekly forecasts. Although $MAD_{M,ijkf}$ decreases when aggregated order data are utilized (360.15 vs. 313.86, $F_{4, 3,355} = 10.31$, $p < 0.01$), it instead increases for aggregated POS data (323.78 vs. 385.97, $F_{4, 3,355} = 9.13$, $p < 0.01$) such that aggregated order data can be a potentially superior demand signal compared with aggregated POS data. Therefore, both H_1 and H_2 are again supported for monthly forecasts, respectively.

Results for $AGG_{M,ijkf} * POS_{M,ijkf} * AF_{M,ijkf}$ show statistical significance as well (see Table 8, Model 3; $\pi_{M,8} = 203.45$, $p < 0.05$). Estimated marginal means obtained are illustrated in Figure 6.

Figure 4: Interaction plot of the impact of autocorrelation on temporally aggregated demand signals and weekly forecast error.

As was observed in the analysis of weekly forecasts, $AF_{M,ijkf}$ has a clear impact on the relationship between temporal approach and MAD. As shown in Figure 6, as $AF_{M,ijkf}$ moves from negative to positive, $MAD_{M,ijkf}$ increases for both demand signals. The positive slope exhibited by both aggregated order and aggregated POS for monthly forecasts offer clear support for H_{3a} and H_{3b} as well.

DISCUSSION

Suppliers require accurate and timely forecasts of their customers' orders to properly position inventory throughout their distribution networks and to schedule outbound logistics operations.

Figure 5: Interaction plot of temporal aggregation and demand signal for monthly forecast error.

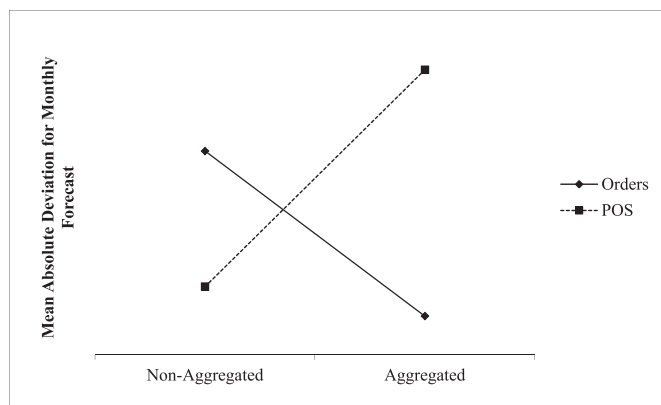
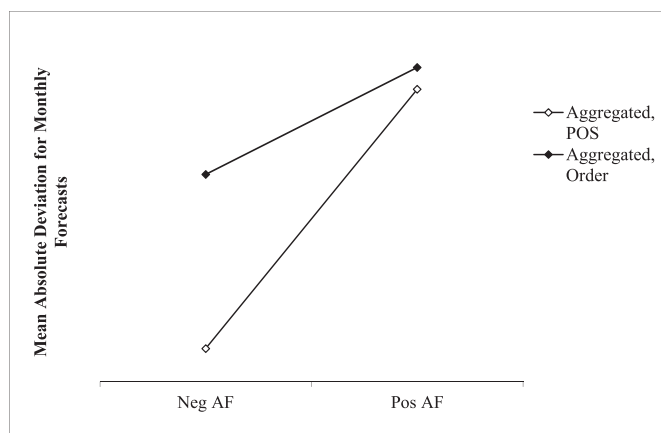


Figure 6: Interaction plot of the impact of autocorrelation on temporally aggregated demand signals and monthly forecast error.



Due to industry consolidation, suppliers are becoming increasingly reliant on utilizing key customer account forecasts (Lapide 2007) to plan for an increasingly higher portion of sales (Hofer et al. 2012). Thus, the timeliness and accuracy of these forecasts are critical.

Customer demand signals obtained from retailer order history are the most immediately available data from which to generate forecasts, however, they are prone to high degrees of fluctuation due to the bullwhip effect (Lee et al. 1997; Croson and Donohue 2002). While one way to mitigate this effect is through POS data sharing, smaller suppliers may not have the resources to implement the necessary infrastructure. Our results suggest that they may instead selectively use a temporally aggregated forecasting approach to take advantage of the variance reduction property of this technique (e.g., when customer order demand signals are characterized by negative autocorrelation).

When a supplier has access to POS data, the prevailing assumption is that it is always a superior source of demand signal, relative to order data, because it tends to have less variance and provides suppliers with a more accurate view of consumer demand (Williams and Waller 2010). As shown, temporal

aggregation may compromise the value of shared POS data due to information loss. However, temporal aggregation's impact on POS is not universally bad. As POS becomes less positively autocorrelated, benefits associated with variance reduction can still eclipse the detriments of information loss.

In summary, temporal data aggregation in forecasting have potential to be a viable strategy to reduce both computational intensity and unpredictable variance, thereby lessening resource requirements as well as increasing forecast accuracy. Furthermore, in conjunction with previous research, the value of shared POS information is shown to be subject to factors such as network complexity in product-location aggregation (Williams and Waller 2011), and autocorrelation factor in temporal aggregation. These conclusions have implications for both theory and practice in logistics and SCM.

With regard to theory, it has been established that temporal aggregation of data results in both information loss (Amemiya and Wu 1972; Rossana and Seater 1995) and variance reduction (e.g., Hotta et al. 2005). Our study makes a contribution to this discussion by first showing that both effects exist concurrently. However, their net impact on forecast error is dependent on the autocorrelation factor of the demand signal prior to aggregation. This effect, in turn, indicates that the temporally aggregated approach (e.g., aggregate data to monthly buckets, forecast in months, divide out forecasts to get weekly projections) is more appropriate for less positively autocorrelated demand signals. Conversely, the temporal aggregation approach should match the planning period (e.g., weekly forecast for weekly plans) for positively autocorrelated demand signals.

MANAGERIAL IMPLICATIONS

This research presents several key findings that provide guidance to managers as they seek to develop more accurate forecasts of retailer DC orders. First, temporal aggregation has distinct impacts on the forecast accuracy achieved with POS and order history data (Figures 3 and 5). In addition, we demonstrate that the net effect of a temporally aggregated forecast approach, using either demand signal (Figures 4 and 6), is largely determined by a relatively simple statistical property that is fairly easy for managers to obtain—the autocorrelation factor. These findings should allow managers to select the best possible forecast approach given the data available.

Our findings also add to the existing body of literature that shows POS to be the superior demand signal for predicting orders from retail customer's DCs. However, we find that temporally aggregating POS data from weekly to monthly can cause the data to lose its predictive superiority over order history data. In fact, when temporally aggregated, our findings suggest there is effectively no difference between the ability of POS and order history data to predict future orders. This is due to the fact that temporally aggregating POS causes information loss, and since there is less variability in POS, there is less variance reduction to be gained.

Therefore, if a supplier has a choice between using POS or order history data to forecast orders, our findings suggest POS is most likely the superior input, and that the supplier should use the POS data in a higher frequency form (weekly) to predict

orders. If the supplier does not have access to POS data and subsequently uses historical order data to predict orders, our findings suggest that the supplier should instead temporally aggregate the data into a lower frequency form (monthly). Doing so will reduce forecast error because, in this case, the variance reduction effect from temporally aggregation will dominate any information loss.

Furthermore, our results indicate that managers must also consider the degree and nature of autocorrelation present in the demand signal data when temporally aggregating. Our findings suggest that a higher degree of positive autocorrelation in the data will lead to increased forecast error when using temporally aggregated POS or historical order data to forecast future orders. This finding is likely due to that the variance reduction benefit via temporal aggregation is lower when either the POS or order history data are positively and highly autocorrelated.

Taken cumulatively, this research indicates that whether managers desire forecasts in weekly or monthly buckets, POS data should be used, if it is available, and should not be aggregated. However, if POS is not available and order history data must instead be used, our research suggests taking the opposite approach; order history data should be temporally aggregated. Finally, managers should measure and consider the degree of autocorrelation in the demand signal being used to forecast orders. The degree of autocorrelation does not influence whether POS or order history data should be used to forecast orders or whether or not to temporally aggregate, but, it does influence how much additional accuracy can be achieved through these choices.

LIMITATIONS AND FUTURE RESEARCH

In this study, we reconciled two conflicting effects of temporal aggregation on forecast error by utilizing paired order history and POS data collected for a large number of SKU-DC combinations. Although analytical literature generally argues that temporal aggregation results in a less accurate forecast, a large number of empirical studies, as well as evidence from industry, point to the contrary. We find that in the retail supply chain, the effect of temporal aggregation on forecast error is dependent on the demand signal. However, we note some limiting factors that should be considered. First, our data come from two high-volume, nonseasonal grocery categories for which the data exhibited a positive trend component. Future research could report on the robustness of our results when other types of product are examined, such as additional grocery categories, nongrocery CPG's, seasonal items, or stable items with neither trend nor seasonality. Furthermore, the products in our sample are fast-moving consumer goods, where the lead times are short compared with products such as electronics or apparel. Thus, the relevant short forecast horizon of 13 weeks or three months may need to be extended for other products. In addition, our data come from a single manufacturer and retailer. As a result, we are unable to assess the potential differences between buyer-supplier relationships types, retail store formats, or retail pricing strategies.

Another limitation of this research involves the fact that there is potential for the autocorrelation factor to change based upon the time frame. Thus, it is conceivable that as the autocorrelation

factor actively changes, the decision as to whether or not to utilize a temporally aggregated forecasting approach may change as well. Future research can examine how dynamic updates to statistical properties of demand signals impact forecasting approach and forecast accuracy. Furthermore, while autocorrelation can theoretically be between -1 and 1 , extreme negative values are rarely observed in practice.

Last, future research could also examine the impact of additional types of forecast methods. Our choice in forecast model is based on our interactions with both practitioners and forecasting software vendors. Hence, we believe that our forecast competition represents settings commonly found in practice. However, with advances in computing power, other, more complex, forecast methods have become increasingly feasible. As detailed by Makridakis and Hibon (2000), highly complex forecast methods make specific assumptions regarding a demand signal's statistical features such as trend, seasonality, and underlying processes. Therefore, challenging any of the commonly held forecast assumptions can be fruitful toward improving demand planning.

McCarthy et al. (2006) described the common managerial approach to forecasting as resembling a "black box." A lack of general understanding in the underlying statistical effects of forecasting continues to impede superior demand planning. In addition, the existence of countervailing theoretical predictions from multiple disciplines hampers effective research in forecasting. This article serves to reconcile those theoretical differences, thereby providing insight and guidance such that the black box might be a little less dark for both industry managers and those conducting forecasting research.

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